

Operating System

Process and Thread management

Process

- * A program in execution is called process
- * It includes
 - ⇒ Program code
 - ⇒ PC → (stores value of next instruction)
 - ⇒ Stack → (local variables, function call)
 - ⇒ Data Section → (global variables)
 - ⇒ Heap → (dynamically allocated memory)

Process States

- * New → created
- * Ready → waiting to be assigned to a processor.
- * Running → instructions are being executed
- * Waiting → waiting for I/O
- * Terminated → execution finished

Process Control Block (PCB)

1. Process ID (PID): A unique identifier for each process
2. Process State: Indicates the current status of process
3. Program Counter: holds address of next instruction
4. CPU Registers: stores process working data (like accumulator, stack pointer) that must be saved during context switch.

5. Scheduling info : Includes priority , queue pointers needed by scheduler to choose next process .
6. Memory management info : about process memory page tables , segment tables etc .
7. I/O Status info : Tracks I/O devices .

Process scheduling

Process scheduling is the one which decides the order in which processes are executed by the CPU

Types of Scheduling

- * Long-Term Scheduling : It is responsible for bringing processes from job queue (secondary memory) to Ready queue (main-memory). It is also known as Job scheduling . It will ensure that there is a good mix of CPU-bound and I/O bound processes.
- * Short-Term Scheduling : It is used to schedule the process from ready-state to running state . It will select the process based on priority or timeslice . It is also known as CPU-scheduling .
- * Middle-term-scheduling : It will enable us to handle swapped out processes . When RAM is full the system pauses some processes & save them to disk and again they are brought back
eg : phone background apps